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SPACE WEATHER-AWARE ANOMALY DETECTION & PROGNOSTICS FOR CUBESATS

A SYSTEMATIC ABLATION STUDY ON FUTURE FUSION STRATEGIES

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ABSTRACT

This study investigates whether fusing space weather indices with CubeSat telemetry improves unsupervised anomaly detection. Four model classes are evaluated under three fusion conditions on ESA's OPS-SAT-AD benchmark. Results show naive fusion degrades all models, while the best telemetry-only approach achieves F1 = 0.604.

BACKGROUND & OBJECTIVES

CubeSat anomaly detectors rely solely on onboard telemetry, blind to external space weather conditions. During Solar Cycle 25's peak, geomagnetic storms cause late detection and elevated false-alarm rates.

Can fusing space weather data with telemetry improve anomaly detection?

DATA & METHODS

CubeSat anomaly detectors rely solely on onboard telemetry, blind to external space weather conditions. During Solar Cycle 25's peak, geomagnetic storms cause late detection and elevated false-alarm rates.

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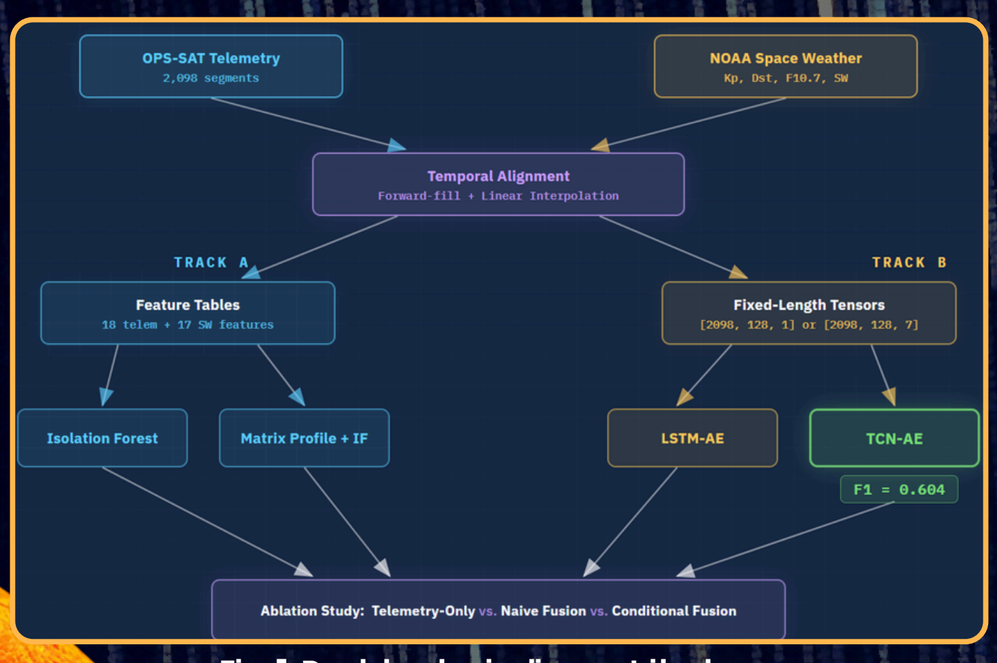


Fig. 1: Dual-track pipeline architecture

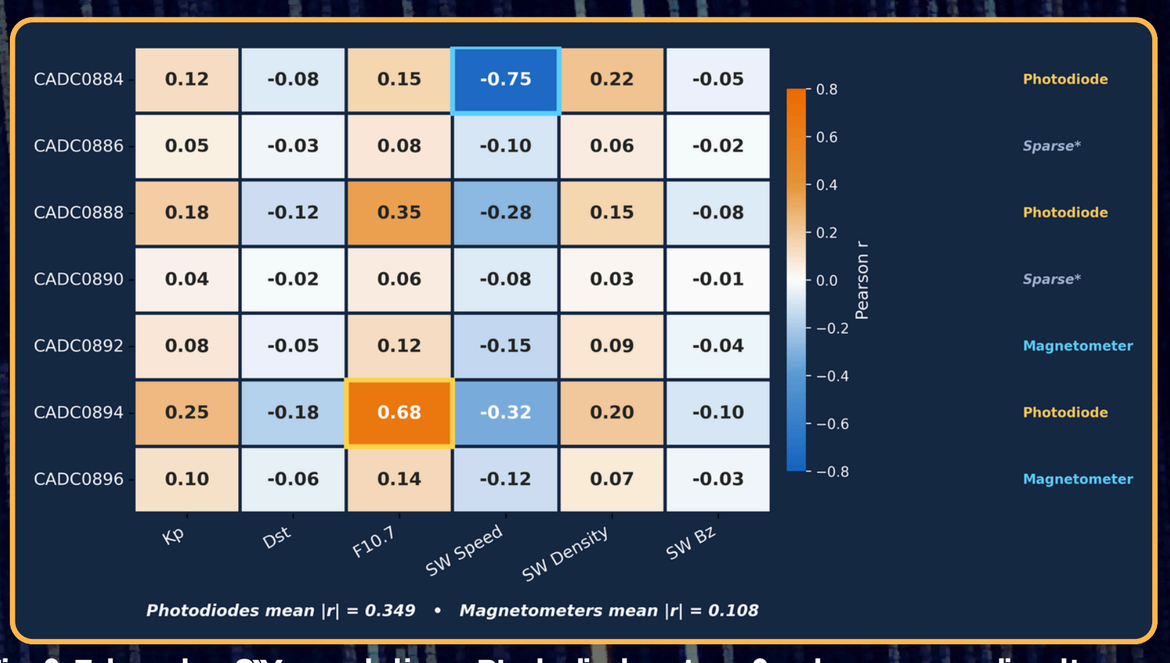


Fig. 2: Telemetry-SW correlations. Photodiodes show 3x stronger coupling than magnetometers (mean |r| = 0.349 vs. 0.108).

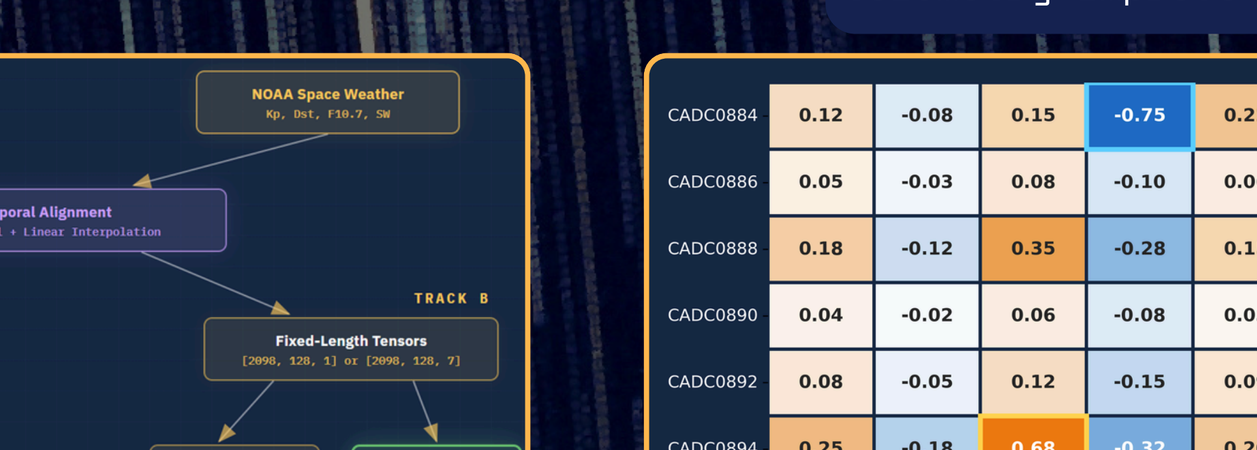


Fig. 3: F1 scores across all models and fusion conditions. Naive fusion degraded every model ($p < 0.01$).

KEY FINDINGS

1. Naive fusion catastrophically degrades all four models (F1 drop: -17% to -63%, $p < 0.01$)
2. Adding SW features as extra reconstruction targets dilutes the anomaly signal
3. Conditional fusion recovers performance but does not surpass telemetry-only baselines
4. TCN-AE telemetry-only achieves F1 = 0.604, the highest unsupervised result on OPS-SAT-AD

CONCLUSIONS & IMPACTS

How you fuse matters more than what you fuse. Naive concatenation is not viable; conditional architectures show promise but require refinement for moderate-storm regimes.

Context-aware detection enables operators to distinguish environmental effects from hardware faults, reducing unnecessary maneuvers and extending CubeSat operational lifetimes.

REFERENCES